Rapid
Online
Assessment of
Reading

NEXT STEPS

Brain Development and Education Lab

Stanford University

School of Medicine

Graduate School of Education

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LEARNING TO READ AT ANY AGE

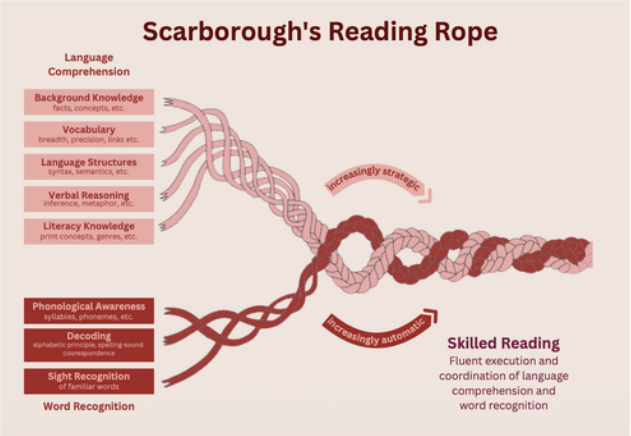
How children learn to read and write is fundamentally different from how children learn to talk and understand speech. While children naturally pick up spoken language from exposure in their environment, learning to read requires literally rewiring the brain to connect sounds, symbols, and meaning. Students must be explicitly taught reading and writing skills. When foundational reading skills are missing, it can impact a child's reading for their entire lives. Fortunately, screening for foundational reading skills can quickly reveal those gaps in the foundation, and reading interventions can close those gaps for readers of any age!

Several decades of reading **research** have shown that there are many word recognition and language comprehension skills that all readers need to master. Effective reading interventions for each skill can be implemented in the **elementary**, **middle**, or **high school** classroom.

In your classroom, you may be working with individuals, small groups, or large classrooms of readers who need support with foundational reading skills. If you are overseeing instruction for many readers, ROAR data can help you identify who needs more support and inform how you **group readers by skills for intervention**.

Effective instruction in foundational reading skills meets students where they are in terms of skill formation with systematic instruction delivered frequently and regularly. Direct instruction accompanied by frequent opportunities for multi-sensory practice and authentic application reinforces these skills. Explaining to readers the science and purpose behind the instructional activities can support readers' motivation!

As an introduction to foundational reading skills, Scarborough's reading rope, shown below, is an excellent depiction of how word recognition and language comprehension skills are woven together to support skilled reading.



UNDERSTANDING THE SCORES

Each ROAR assessment evaluates foundational reading skills that are crucial for your student's literacy development. The specific assessments your student takes will be chosen by the school. These are likely to include one or more of our four key assessments that evaluate your student’s skills in letter names and sounds, phonemic awareness, word decoding, and sentence reading efficiency. Scores that fall into the Developing Skill and Needs Extra Support categories may indicate specific areas of focus that your student needs support in.

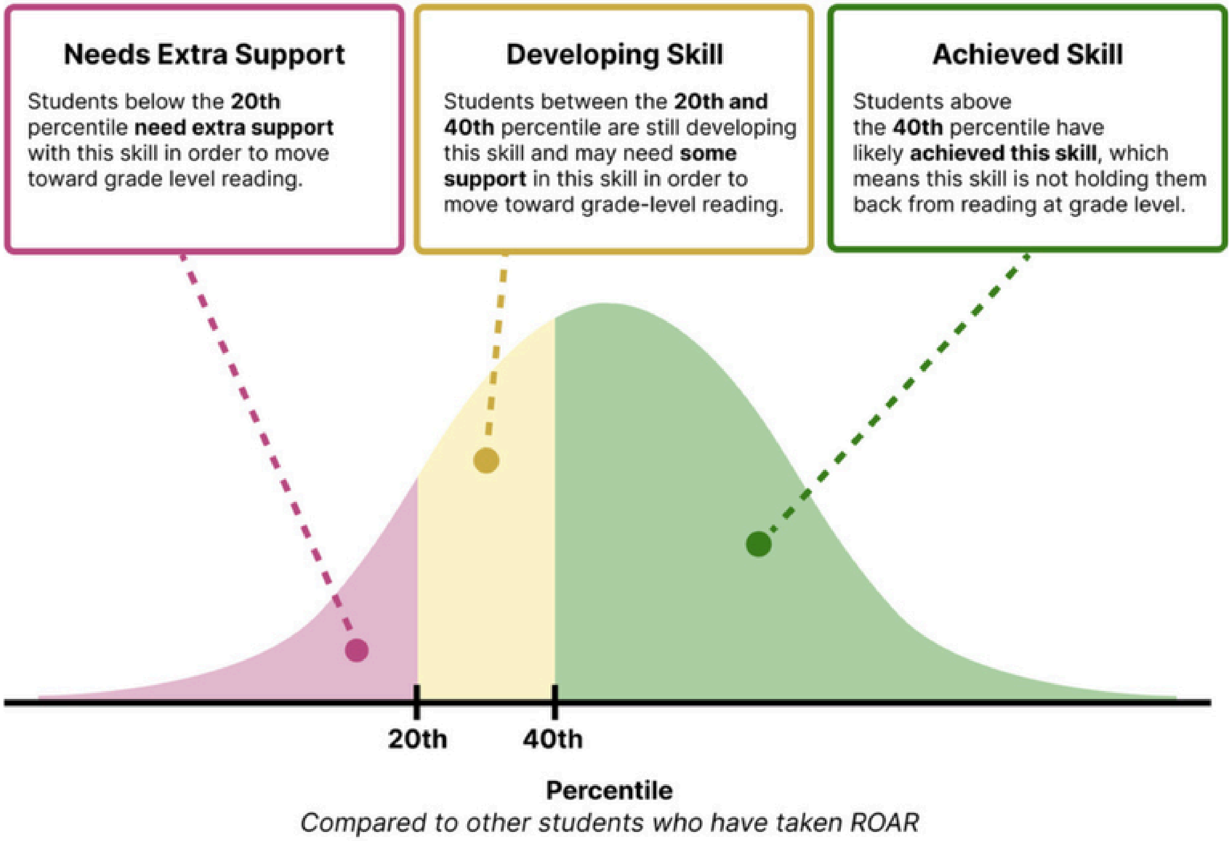
Depending on the assessments your student has completed you may be able to view their scores as a percentile, standard score, or raw score.

The ROAR assessments return four types of scores:

- **Raw score:** This score captures your student’s general performance on the assessment, such as total items correct. This score is difficult to interpret on its own which is why it is used to generate standard scores and percentile.
- **Standard score:** This score compares your student’s score to the performance of other students in their age or grade group. This score gives you a glimpse of your student’s understanding of the tested skill compared to their peers.
- **Percentile:** This score also compares your student’s score to all the students in the same grade who have taken ROAR. If students were ranked from the lowest to highest score, this score would be your student’s ranking. For example, if your student is in the 74th percentile, then they scored better than 74 out of 100 students.
- **Support Category:** This score can help identify students in the most urgent need of intervention by providing a color (pink, yellow, green) associated with their score’s percentile. For elementary students, this score is determined by the 20th and 40th percentiles and for secondary students, this score is based 3rd and 5th grade percentiles. Details on the thresholds for this score type are included below.

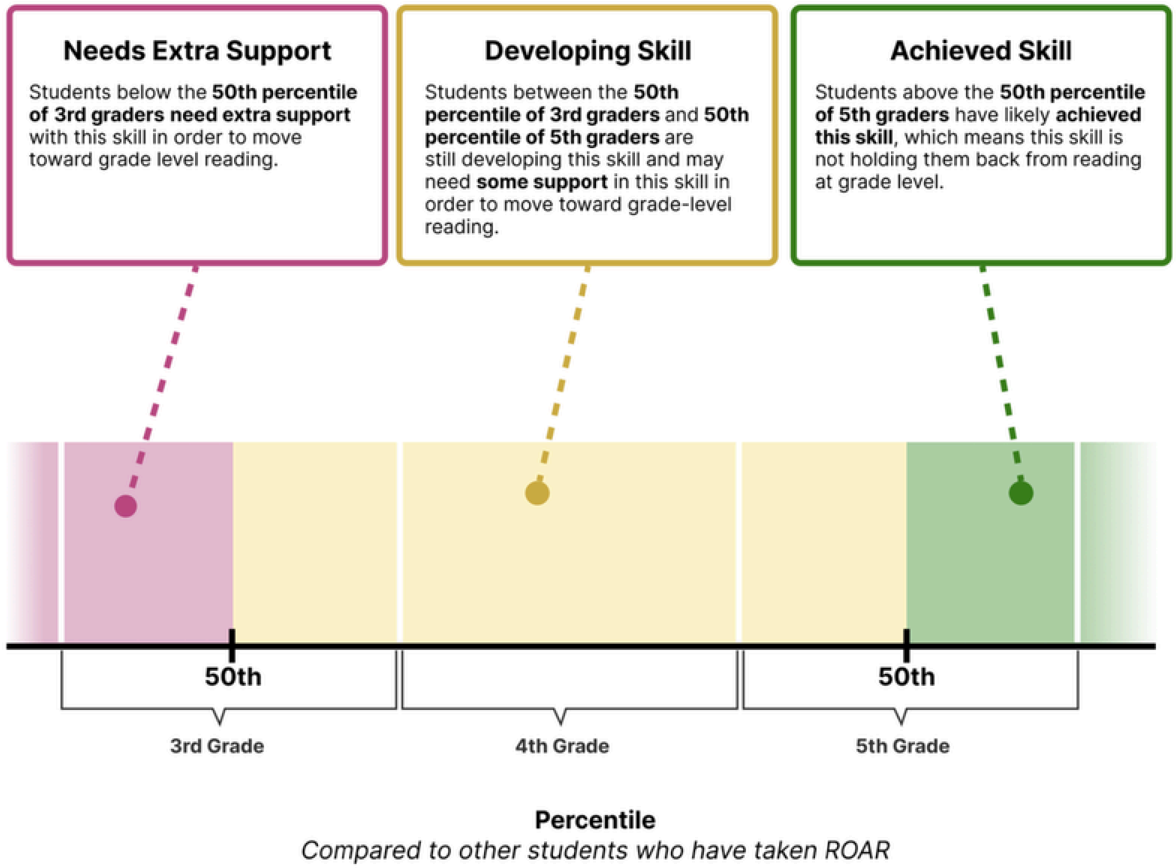
Learn More About Standard Scores and Percentiles:
Southeast Psych standard scores video explanation

ELEMENTARY SCORE REPORT (GRADES K-5)



Needs Extra Support	This category indicates that your student's score is below the 20th percentile for their age group. This assessment suggests that **this skill is holding them back from accessing grade-level material.** They will likely benefit from additional systematic, intensive instruction on this skill in addition to grade-level instruction in order to access grade-level material.
Developing Skill	This category indicates that your student's score is between the 20th and 40th percentile for their age group. This means that their skill level may be below average compared to most students in their age group. Your student will continue to grow in this skill as they read in their classroom. This assessment suggests that **this skill may need focused intervention** in addition to grade-level instruction in order to access grade-level material.
Achieved Skill	This category indicates that your student's score is above the 40th percentile for their age group. Your student will continue to grow in this skill as they read grade-level material in their classroom with rich Tier 1 instruction. This assessment suggests that **this skill is not holding them back from accessing grade-level material.**

SECONDARY SCORE REPORT (GRADES 6-12)



For students **in grade 6 and above**, support level categories are based off of **3rd - 5th grade percentiles** as those are the grades in which foundational reading skills are typically mastered.

Needs Extra Support	This category indicates that your student's score is at or below a 3rd-grade level with this skill. Students with this support level **likely need additional systematic, intensive instruction on this skill** in order to access grade-level reading.
Developing Skill	This category indicates that your student's score is between a 3rd and 5th grade level with this skill. Students with this support level **may need additional instruction on this skill** in order to access grade-level reading.
Achieved Skill	This category indicates that your student's score is above a 5th grade level. Students who have achieved this skill **likely do not require intervention on this skill** to access grade-level reading.

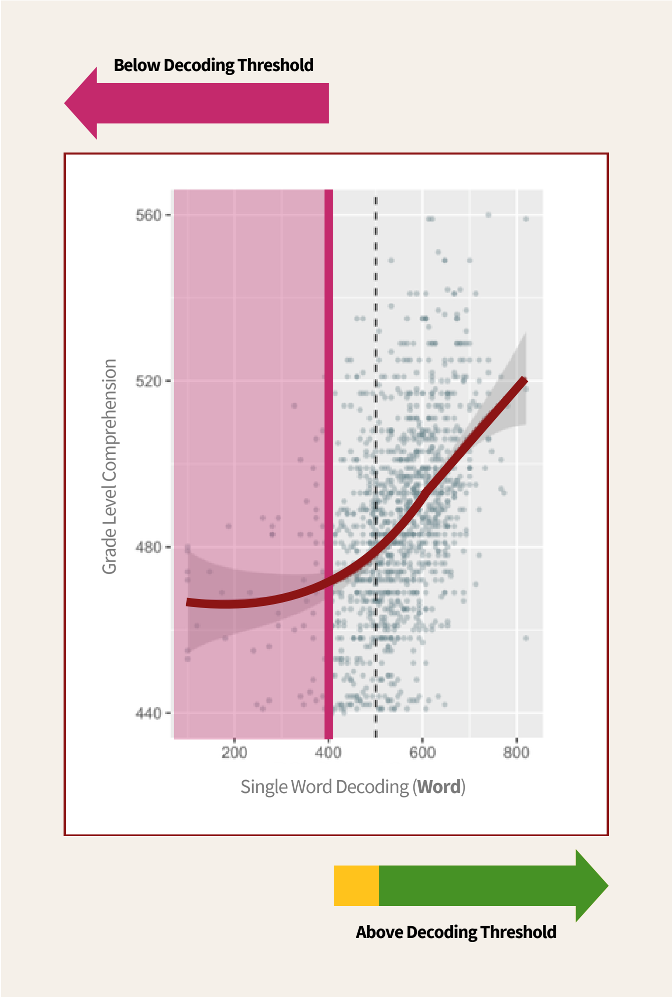
SECONDARY SCORE REPORT (GRADES 6-12)

DECODING THRESHOLD

As students develop their foundational reading skills we often expect their "grade-level" reading achievement to develop at similar rate as decoding. However, researchers and educators alike don't always see this relationship. In fact, there seems to be a **decoding threshold** that students must pass for their decoding skills and grade-level achievement to improve at similar rates.

The graph shows the relationship between **single-word decoding skills** and **grade-level achievement.** From left to right, decoding skills increase. From bottom to top, grade-level achievement increases. **The pink vertical line marks the decoding threshold.**

Above the decoding threshold, stronger decoding skills are closely associated with higher grade-level achievement, as shown by the upward slope of the red line. **Below the decoding threshold,** decoding skills are less strongly related to grade-level achievement, as shown by the relatively flat red line in the pink shaded area.



If a secondary student's score on Word receives the pink support category, their decoding skills are below the decoding threshold and they would benefit from **targeted phonics support and intervention** to access grade-level reading. If a secondary student's score on Word receives the green support category, decoding is not likely a skill holding them back from grade level reading. If they are demonstrating difficulty with grade-level material, they likely need support in another area, such as their language comprehension skills.

DATA-BASED DECISION MAKING

QUESTIONS TO CONSIDER AT THE INDIVIDUAL LEVEL

While interpretting ROAR scores and analyzing the skills students need support with, it is important contextualize assessment results with a learner's unique circumstances. Consideration of the following questions may help guide decisions for supporting individual learners.

Does the student's responses likely reflect their best effort?
We have made the assessment child-friendly and engaging, but other factors may contribute to a child's effort in a given day. Consider their comfort, motivation, and other factors that may have come into play, affecting their ability to showcase their true skill set on any measure. The ROAR Score Reports also provide more detail on this point by flagging scores as "unreliable" when a student speeds through the assessment.

Was the student's assessment environment distraction-free, setting them up for success?
ROAR is designed to be fast and engaging, and it provides many opportunities for young students to take breaks without impacting scores. However it is, of course, important to consider aspects of the environment that might contribute to differences in performance.

If the student was clearly not engaged and giving their best effort, is this indicated in the score report?
The ROAR score report automatically flags students' scores when students are displaying signs of disengagement and not providing a reliable metric of their true ability (e.g., clicking too fast and/or repetitively). These are marked as unreliable and educators can easily filter for these scores via the sort and filter features on the online ROAR dashboard.

Does the child have a 504 plan or an Individual Education Plan (IEP)? Are there specific accommodations and considerations surrounding the student's special education that should be considered?
If a student has a 504 plan or an IEP, the guidelines provided in the ROAR Guidance for Students Receiving Special Education Services and 504 Accommodations should be considered and followed.

What is the student's prior education?
How will this affect their current foundational reading skills abilities? While screening for foundational reading skills is important and informative for making instructional decisions in early Kindergarten, students should experience substantial, consistent, systematic, structured instruction in foundational reading skills prior to considering a low reading score as a risk factor for dyslexia. Educators should allot time and provide materials from a high-quality curriculum for students to practice the foundational reading skills that follows a scope and sequence prior to considering a low reading score as a risk factor for dyslexia in Kindergarten.

Is the student proficient in English? Are there any accommodations and considerations surrounding the student's language proficiency and multilingualism?
The ROAR Guidance for English Language Learners helps educators consider the appropriate language of assessment and interpret scores for multilingual learners. If a student is a fluent Spanish-speaker, the student may be assessed in their home language with the ROAR-Español measures. If students are scoring high in their home language, the evidence from ROAR would suggest they do not have issues with foundational reading skills, so support can focus on supporting further language development in English and Spanish. If the student is scoring low in both languages, support in foundational reading skills should accompany other support with language development. Vignettes in the guide linked above provide further nuance.

QUESTIONS TO CONSIDER AT THE GROUP LEVEL

Another step in this conversation is to review at the group level. Educators often have to decide the most effective distribution of limited resources for high impact improvement of literacy. Consideration of the following questions may help guide these decisions.

What percentage of the class needs extra support?
The ROAR score report displays a pie chart which separates the proportion of students who are in each support category for each of the foundational reading skills assessments. Educators can hover over these pie charts to identify the specific count of students within each support category as well. This visualization allows educators to quickly identify the percentage of the class that needs extra support in a given skill. Additionally, the ROAR score report provides a table of all students' individual support categories for each of the foundational reading skills. The educators can easily sort and filter the table to identify which specific students fall into the "needs extra support" category.

Do some students need help with all skills, while others need focused support in one?
Educators can quickly identify which students need support on the foundational skills by glancing at the student table mentioned above. The ROAR score report also provides sub-skill tables for ROAR-Letter and ROAR-Phoneme. The ROAR-Letter table highlights which letter names and sounds each student needs improvement on and the scores for each sub-skill within the assessment (i.e., lowercase and uppercase names and letter sounds). The ROAR-Phoneme table indicates the separate scores for each sub-skill within the assessment (i.e., first sound matching, last sound matching, and deletion) and shows which sub-skills each individual student needs to work on. These features help inform educators on the specifics of each student and the larger organization as well.

What should be done at the whole class level, small group, or individual level?
ROAR often identifies a gap for a whole class - for example consider the scenario where a majority of the class is struggling in Phonological Awareness. In this scenario, the teacher should focus more class time on this skill. In addition to using group level data to inform changes to the general instructional approach, it is still important that individual students receive additional support through a well-designed multi-tiered system of support as outlined above. For example, even in the scenario where a teacher identifies Phonological Awareness as an instructional focus for the class, an individual student that is in the "needs extra support" category on all the measures should still be moved to a more intensive tier of instruction.

RECOMMENDATIONS FOR STUDENTS WHO NEED EXTRA SUPPORT

As detailed in the Professional Development resources available on the ROAR webpage at roar.stanford.edu, we use the term “needs extra support,” rather than “high-risk,” because it is a more accurate description of the action the assessment should lead to. The teacher and instructional community around the student need to incorporate additional contextual information to determine the specific supports that will be most impactful for the student, and ROAR assessments provide recommendations.

If a student is determined to “need extra support,” the first step in this conversation is to review the developmental, environmental, linguistic, and instructional context for this student to determine which next steps to take.

While there are several exceptions detailed in this guide, based on extensive analysis, we have arrived at these recommendations:

1. If a student is reported to be in the category “needs extra support” or “developing skill” in any required skill, they should be provided with additional, direct instructional support in that skill in small groups or individually.

2. If a student is reported to be in the category “needs extra support” on **two or more required assessments**, their instruction should be moved to a higher tier of support with a more focused instructional approach (i.e. Tier 2 or Tier 3). Here they should receive frequent, high-quality, structured, systematic instruction and practice following a scope and sequence, and using an evidence-based program, in a small group or individual setting for at least three months. Frequent formative assessment of a student’s progress will inform instructional adjustments and differentiation.

3. They should be re-assessed using the ROAR Foundational Reading Skills Suite at the end of this period. If at this time they still score in the category “needs extra support,” on two or more required assessments, they are at-risk for dyslexia, and they should again move to a higher tier of support for intensive, systematic instruction. At this time, more comprehensive assessment can provide a detailed profile of strengths and weaknesses for supporting targeted intervention.

    - In the ideal case, the student who has not seen growth in Tier 2 support will begin receiving Tier 3 support that includes one-on-one daily instruction with a highly-qualified teacher (or interventionist) who is trained in the appropriate instructional methods for teaching foundational reading skills at the child’s developmental level. This is the approach most likely to result in long-lasting results. However, if they made substantial growth in Tier 2 support in that initial period, they may stay at that level of support, since it is working for them.

- Consider, for example, a first grader who is screened at the beginning of the year, and it is indicated that they need extra support based on scoring below the 5th percentile in ROAR-Word and ROAR-Sentence. After they receive the recommended targeted support in word-level decoding and reading fluency, they score in the 15th percentiles in the winter. They will still be in the pink “needs extra support” category on both assessments, but their substantial progress will indicate that the current intervention is working for them, and they should continue.

4. This decision should be reassessed at each testing window, in the fall, winter, and spring.

FOUNDATIONAL READING SKILLS

Decades of reading research have resulted in numerous models of reading development. However, several key ideas remain consistent throughout the literature. Reading comprehension requires both efficient recognition of words and understanding language— both abilities which depend on underlying skills that develop over time and interact to support increasingly complex reading.

Foundational skills such as phonological awareness, letter–sound knowledge, decoding, and fluency support accurate and increasingly automatic word reading, while vocabulary, language knowledge, syntax, and inferential reasoning support understanding. As readers develop, these skills become increasingly interconnected, allowing more of their attention to shift toward constructing meaning from written text.

SOUNDS → **WORDS** → **SENTENCES** → **MORE COMPLEX TEXTS**

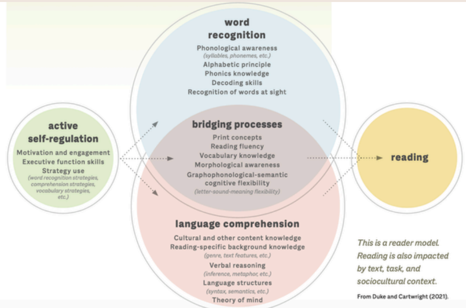
Mastery of these skills improves...

Mastery of these skills improves...

Mastery of these skills improves...

COMMON MODELS OF READING DEVELOPMENT

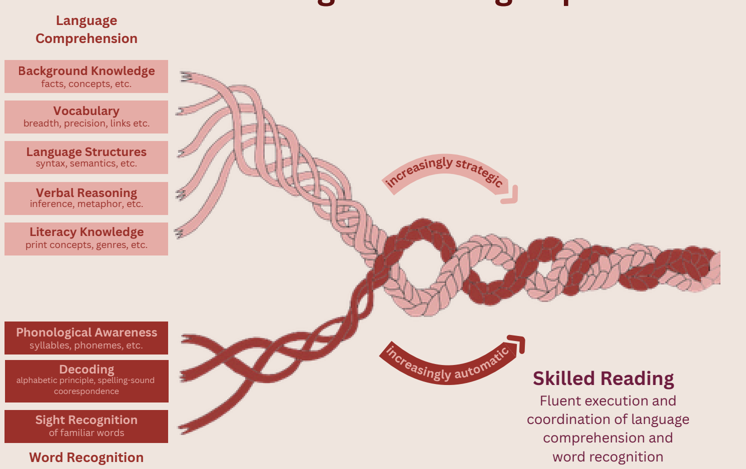
Active View of Reading



Simple View of Reading

D x LC = RC

Scarborough's Reading Rope

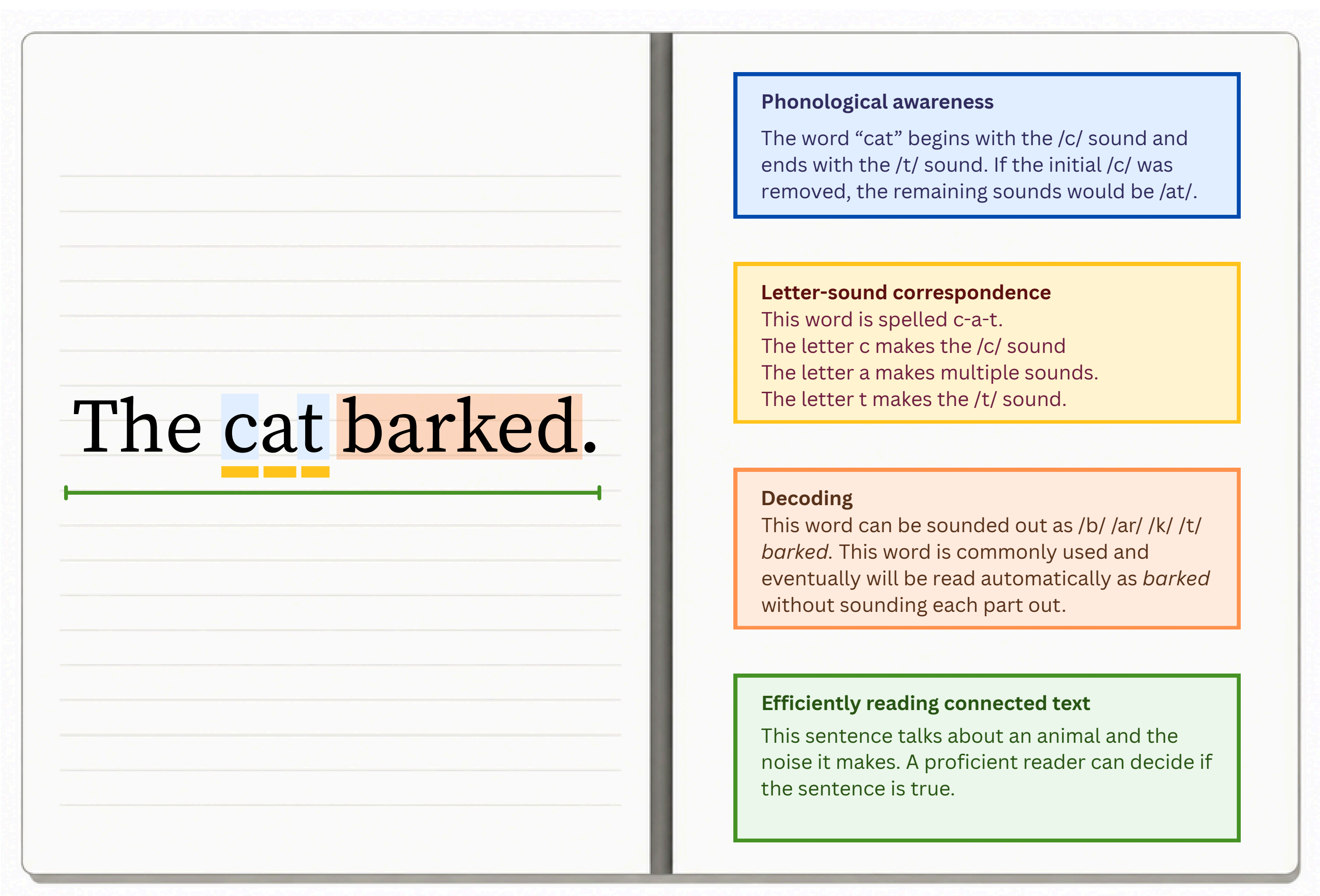


Duke & Cartright, 2021; Gough & Tumner, 1986; Mitchell, 2026; Scarborough, 2001; Serravallo, 2023; UFLI, 2021

FOUNDATIONAL TEXT LEVEL READING SKILLS

Foundational word recognition skills refers to a students ability to recognize and read written words accurately and efficiently. They include the ability to recognize and manipulate the sounds in spoken words, connect letters and letter patterns to those sounds, and recognize familiar words automatically. These skills work together to support increasingly efficient and fluent reading at the sentence level and, eventually, within complex texts.

As students become more proficient readers, word recognition becomes increasingly automatic, allowing them to devote more attention to constructing meaning from text.



FOUNDATIONAL TEXT LEVEL READING SKILLS

PHONOLOGICAL AWARENESS

Phonological awareness is the ability to recognize and manipulate phonemes, or the sounds of spoken language. This is fundamental to developing reading skills.

Assessed by ROAR-Phoneme

LETTER-SOUND KNOWLEDGE

Letter-sound knowledege is the ability to connect phonemes, the sounds of spoken language, to symbols (the alphabet). This skill must be increasngly automatic to support the rapid decoding of unfamiliar words.

Assessed by ROAR-Letter

WORD RECOGNITION + DECODING

Decoding is the act of mapping sounds to letters and words. Decoding is the first reading skill and is the foundation of rapid and automatic word recognition.

Assessed by ROAR-Word

READING EFFICIENCY

Reading efficiently means fluently reading connected text at an efficient rate. This is important for comprehending increasingly complex text.

Assessed by ROAR-Sentence

ROAR-PHONEME

PHONOLOGICAL AWARENESS

WHAT IS IT?

Phonological awareness is the ability to recognize and manipulate *phonemes*, or the individual *sounds* of spoken language. While it does not involve written language, the development of phonological awareness is a critical foundation of literacy. Phonological awareness skills include tasks like segmenting words into sounds, blending sounds to form words, and manipulating sounds within words.

WHY IS IT IMPORTANT?

Phonemes are the building blocks of spoken language just as letters are the building blocks of written language - a young reader's challenge is building the connections between the two. Recognizing and manipulating the sounds in words is the foundation of the ability to connect sounds to letters and syllables, which is critical for learning to decode increasingly complex words. Down the road, it is important in helping children learn new words they encounter in print.

HOW DO READERS DEVELOP THIS SKILL?

Many children pick up on the distinct sounds that make up words by being immersed in a speech-rich environment. The process of being taught to read with an emphasis on phonics also stimulates the development of phonological awareness for **some children**. However **many others** need more support. Games and other activities that draw attention to the individual sounds and syllables in words give your reader more targeted practice in this skill. Practicing recognition and manipulation of sounds in words can be done with readers of all ages both in the classroom and at home. Because Phonological awareness instruction complements phonics instruction, many programs work on both skills in the same lesson.

NEXT STEPS

A phonological awareness training program should be structured and systematic, ideally following a scope and sequence. For more information and specific activities, check out the following resources:

For Grades K-5:
Florida Center for Reading Research

For in-depth information including tips for all readers:
Review of Scientific Reading Research for Teachers

ROAR-LETTER

LETTER-SOUND CORRESPONDENCE

WHAT IS IT?

Letter-Sound Correspondence is the ability to map sounds to letters, and letters to sounds. It is a foundational skill for learning to read and spell, as it allows students to connect the sounds they hear in spoken language with the letters that represent those sounds in written language. Letter-sound correspondence skills include identifying the sounds associated with individual letters, recognizing letters that represent particular sounds, and applying these relationships to decode unfamiliar words.

WHY IS IT IMPORTANT?

Mapping letters to sounds is the foundation of decoding skills. Children need systematic, direct instruction and practice to master letter-sound correspondences and apply these rules to decoding, beginning with short simple words and progressing towards more complex spelling patterns. Some children pick up letter sound knowledge quickly. Many others require extensive practice to achieve mastery. It is best to teach letter-sound correspondence from the beginning when teaching children about the letters.

WHAT ARE THE COMMON CHALLENGES WITH THIS SKILL?

- Readers may learn the names of letters without learning the sounds the letters represent.
- Readers may learn one sound a letter can make but miss other sounds, such as knowing *hard c* (the first sound in “cat”) but needing to learn *soft c* (the first sound in “city”). English is an opaque orthography, meaning that the correspondence between letters and sounds is more complex than transparent orthographies like Italian and Spanish. This complexity makes English more difficult to learn, particularly at the beginning, but research shows that mastering this complexity is important for long-term success.
- Readers often have an easier time with long vowels than short vowels. Certain letters are easily confused by young readers (e.g., b, d), as are certain phonemes. Noting these errors and providing continual practice and scaffolding until the reader has internalized them is important for the development of later, more complex skills.

HOW DO READERS DEVELOP THIS SKILL?

Children need systematic, direct instruction and practice to master letter-sound correspondences and apply these rules to decoding, beginning with short simple words and progressing towards more complex spelling patterns. Some children pick up letter sound knowledge quickly. Many others require extensive practice to achieve mastery. It is best to teach letter-sound correspondence from the beginning when teaching children about the letters.

NEXT STEPS

A phonics program should be structured and systematic, following a scope and sequence. For more information and specific phonics activities, check out the following resources:

For Grades K-5:
Florida Center for Reading Research

For grades 3+:
Check out Open Source Phonics

For in-depth information including tips for all readers:
Review of Scientific Reading Research for Teachers

ROAR-WORD

SINGLE WORD RECOGNITION

WHAT IS IT?

Two skills are essential for success in reading individual words: Decoding and Automaticity. People with dyslexia struggle with both. **Decoding** is the ability to translate a word from print to speech using the knowledge of how letters represent sounds. Basically, it is the ability to sound out a word accurately. **Automaticity** is the fast, effortless word recognition of words "by sight." These skills go hand in hand: decoding knowledge supports the development of automaticity in word reading.

WHY IS IT IMPORTANT?

Decoding words accurately and quickly is the foundation of reading fluency and comprehension. Learning to decode more complex words unlocks more complex texts. Supporting a reader with decoding skills is the key to supporting most chronically struggling readers, and it is possible at any age. Ideally decoding skills are established early in elementary school but interventions are still highly effective at any age.

HOW DO READERS DEVELOP THIS SKILL?

Readers practice decoding letters, syllables, and words into sounds through direct instruction and regular practice. Decoding skills do not develop naturally through immersion, the way spoken language is developed. Direct, explicit, systematic phonics instruction that follows a defined scope and sequence of letter-sound relationships has been shown to be more successful in developing word reading compared to teaching letter-sound relationships as they are encountered in books in a random, unplanned order. It is also important to keep in mind that **people vary dramatically** in the ease with which they can apply letter-sound knowledge to decoding. Some students require light scaffolding while many others require continued, structured, intensive instruction and practice. This variability is why word reading assessments are so critical!

NEXT STEPS

Students who score "need extra support" on Word would benefit from direct, explicit, systematic phonics instruction that follows a defined scope and sequence.

Provide opportunities for readers to practice reading short words: Begin with simple consonant-vowel-consonant (CVC), and systematically building up to complex, multisyllabic words. Remember English is a very opaque orthography with many exceptions to every rule - this presents a serious challenge for most students. A reading program should be structured and systematic, following a scope and sequence and students will vary in the amount of time they need to master each step in the sequence.

For more information and curricular support, check out the following resources:
For Grades K-5: Florida Center for Reading Research Activities
For Grades 4-9: What Works Clearinghouse guide to reading interventions
For grades 3+: Open Source Phonics provides free phonics lessons accompanied by decodable texts
For all grades: UFLI Toolbox is an inexpensive phonics program from University of Florida Literacy Institute.

ROAR-SENTENCE

SENTENCE READING EFFICIENCY

WHAT IS IT?

Reading fluency and efficiency refers to the speed at which students can accurately read and understand connected text. Beyond accurately decoding individual words, students vary substantially in the rate at which they can read words and sentences. This is a common limiting factor as students progress from learning to read to reading to learn.

WHY IS IT IMPORTANT?

The goal of literacy is to unlock a new form of communication through written language. But if reading is slow and effortful then it presents a barrier to comprehension particularly as the text a student is expected to read grows longer and more complex. Developing reading fluency and efficiency makes accessing information through written language as natural as spoken language. Readers who are reading slowly or inaccurately at the sentence-level miss out on key connections across the text. Reading remains difficult and unenjoyable, and comprehension is a challenge, particularly with time constraints.

WHAT ARE THE COMMON CHALLENGES WITH THIS SKILL?

- Particularly for children with dyslexia, reading fluency and efficiency can remain a challenge even as they develop decoding skills.

- Mastering decoding skills is important for building fluency and efficiency. Students who miss this foundation struggle to ever establish fluency.

HOW DO READERS DEVELOP THIS SKILL?

- Efficiently reading sentences depends on decoding skills and, particularly, on developing automaticity in decoding.

- For many students, reading efficiency comes with practice after they've established a strong foundation in decoding. For students with dyslexia, reading efficiency can remain a barrier and requires intensive and targeted intervention.

- Repeated Reading has been identified by research as a key strategy for developing fluency. It has two essential elements: 1) reading and re-reading the same text, 2) oral reading with corrective feedback.

- Direct instruction and practice in reading with prosody, or with expression, is a fun and evidence-based way to improve reading efficiency. However students with dyslexia will need more intensive support.

- Fluency intervention for paragraphs and passages should include modeling, repeated reading, and feedback. Choral reading and systematic progress monitoring for fluency also have research support.

NEXT STEPS

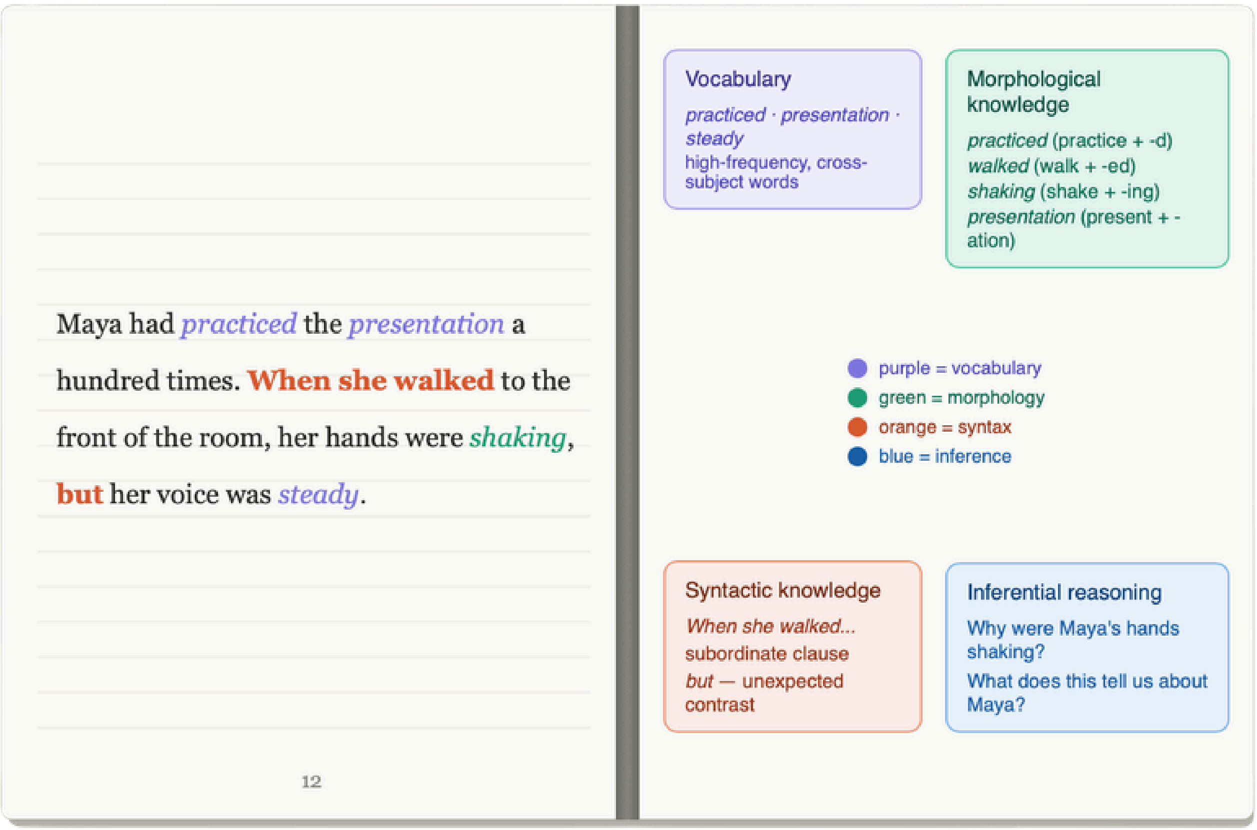
A reading program should be structured and systematic, following a scope and sequence that builds efficiency, fluency and comprehension on a strong foundation of decoding skills. For more information and specific sentence reading activities, check out the following resources:

For Grades K-5: Florida Center for Reading Research Activities
For Grades 4-9: What Works Clearinghouse guide to reading interventions

LANGUAGE COMPREHENSION SKILLS

Foundational reading skills are necessary but not sufficient for understanding text. Language comprehension refers to the ability to construct meaning from what is read. It depends on the language knowledge a reader brings to the page: knowledge of words, word structure, sentence grammar, and the ability to reason across ideas.

These skills develop and interact across the school years. The relative contribution of each may shift as students become more proficient readers, but all four matter for supporting students' ability to understand increasingly complex texts across content areas.



SYNTAX

Syntactic knowledge is understanding how words combine to form meaningful sentences — including how word parts, word order, phrases, and grammatical structures convey meaning. Readers who struggle with syntax often miss meaning even when they can decode every word.

Assessed by ROAR-Picture Syntax

VOCABULARY

Vocabulary is knowledge of what words mean — both individually and in context. Readers with larger vocabularies can access more complex texts and better understand what they read.

Assessed by ROAR-Written Vocabulary

MORPHOLOGY

Morphological knowledge is the ability to recognize and use meaningful word parts — prefixes, suffixes, and roots. It supports vocabulary, decoding, and comprehension of written language.

Assessed by ROAR-Written Morphology

INFERENCE

Inferential reasoning is the ability to connect ideas from text and with background knowledge to understand meaning that is not stated directly. It is a hallmark of skilled, deep reading.

Assessed by ROAR-Written Inference

ROAR-PICTURE SYNTAX

SYNTACTIC KNOWLEDGE

WHAT IS IT?

Syntactic knowledge is understanding how words combine to form meaningful sentences, how word order, phrases, and grammatical structures work together to convey meaning. When a reader encounters "*The cat was chased by the dog,*" syntactic knowledge is what tells them the cat is being chased, not doing the chasing, even though the cat comes first in the sentence.

ROAR-Picture Syntax measures this understanding through listening: students hear a sentence through headphones and select the picture that best matches its meaning from four choices. Because the assessment is delivered through audio, it is appropriate for all students (including those who are still developing their reading skills) and captures language comprehension directly, without the confound of decoding or reading connected text demands.

WHY IS IT IMPORTANT?

Reading requires more than recognizing individual words. It requires understanding how those words work together in phrases and connected text. Sentences in academic texts often have complex structures: passive voice, embedded clauses, multiple modifiers, subordinating phrases. A student who struggles with these structures may decode every word correctly, but still miss the meaning of what they have read.

Research consistently shows that syntactic knowledge contributes to language comprehension alongside vocabulary and decoding and that this relationship grows stronger as students encounter more complex academic texts across the grades. Importantly, syntactic difficulties are often invisible without assessment: a student may appear to be reading fluently by stringing words together efficiently while missing meaning at the sentence level. ROAR-Syntax provides an efficient way to identify students who may need targeted support in this often-overlooked area.

WHAT ARE THE COMMON CHALLENGES WITH THIS SKILL?

- Sentences where function words, rather than position in the sentence, indicate who is doing what — such as passive constructions (e.g., "*The ball was kicked by the girl*") — are harder to process than active sentences, and students often default to the first noun as the actor.

- Complex sentences with embedded clauses, multiple phrases, or subordinating words (e.g., "*although*," "*unless*," "*while*") place higher demands on working memory and can cause students to lose track of meaning mid-sentence.

- Academic texts use syntactic structures that rarely appear in everyday conversation — students may have limited exposure to these patterns outside of school reading, which makes them harder to process automatically.

- For English language learners, syntactic patterns in English may differ significantly from those in their home language, creating specific challenges with structures like passive voice or relative clauses.

- Some students show persistent difficulties with complex syntactic structures even as their vocabulary and decoding skills grow. Many of these students are not identified for support.

HOW DO READERS DEVELOP THIS SKILL?

Syntactic knowledge develops progressively from preschool through secondary school, with students mastering increasingly complex structures over time. Simple subject-verb-object sentences come first (e.g., "*I like ice cream*"); relative clauses, passive constructions, embedded phrases, and structures common in academic writing develop later and continue growing into adolescence (e.g., "*the ice cream that is sitting on the counter is melting*").

All students benefit from rich exposure to oral and written language alongside explicit instruction in sentence structure. Students who show persistent difficulty with complex syntax benefit most from intensive, targeted instruction — teaching specific structures directly, providing repeated practice with varied examples, and helping students notice how grammatical cues signal meaning. Instruction that connects sentence structure to comprehension, rather than treating grammar as an isolated skill, produces the strongest outcomes.

The difference between "*The dog chased the cat*" and "*The cat was chased by the dog*" is purely syntactic — same words, same event, different structure. A reader who cannot reliably process passive constructions will misread the second sentence, and may never realize it.

NEXT STEPS

Syntactic knowledge responds to explicit, targeted instruction — particularly for students who struggle with complex sentence structures. The following programs and resources support syntactic development across grades:

For grades 3–5:
- CLAVES — a language-based literacy curriculum supporting syntactic and vocabulary development, with evidence for English learners and their peers.

For all grades:
- Reading Universe — practitioner resources on syntax and reading comprehension.

- Keys to Literacy — practitioner resources on teaching syntactic awareness and sentence structure.

- Florida Center for Reading Research — free classroom activities targeting sentence construction, word order, and understanding connectives.

- Think SRSD — sentence-level writing and comprehension strategies with research support.

ROAR-WRITTEN VOCAB

VOCABULARY KNOWLEDGE IN CONTEXT

WHAT IS IT?

Vocabulary knowledge is understanding what words mean, not just knowing that a word exists, but being able to access its meaning quickly and reliably when you encounter it in text. ROAR-Written Vocabulary focuses specifically on *core vocabulary*: the high-frequency words that appear consistently across subject areas and grade levels, words like *analyze, conclude, evident,* and *adequate* (Hiebert et al., 2018). These words form the backbone of the texts students read across all content areas.

In this assessment, students read sentences containing a target academic word and select the best synonym from four choices. This format assesses vocabulary knowledge in context, where meaning must be accessed quickly and reliably to support comprehension. Proficient readers will actively and efficiently use their vocabulary knowledge in concert with background, text structure, and morphological knowledge to glean meaning from text.

WHY IS IT IMPORTANT?

Vocabulary knowledge is one of the strongest predictors of reading comprehension. When students know the words in a text, they can focus their attention on understanding ideas, making connections, and reasoning across the passage. When they do not, they expend cognitive effort trying to figure out word meanings, which can detract from comprehension. Research suggests that readers need to know approximately 98% of the words in a text to comprehend it without support.

Core vocabulary is particularly important because these words appear across subject areas. A student who knows what *analyze*, *compare*, and *conclude* mean has tools that work across content areas in science, social studies, math, and English — for example, a student who knows that *analyze* means to examine something carefully and break it into parts can apply that knowledge in science lab reports, history essays, and math problem explanations. As texts become more complex in the upper elementary grades and beyond, fluent knowledge of core vocabulary becomes an increasingly critical foundation for reading success. Unlike highly specialized content words (for example, a technical vocabulary item like "iris" from science) core vocabulary words are the words that cut across all subjects and are rarely taught directly.

WHAT ARE THE COMMON CHALLENGES WITH THIS SKILL?

- Students may only encounter academic vocabulary words in reading without ever receiving explicit instruction in their meanings. Unlike everyday words, these words rarely come up in casual conversation. Students build exposure to them primarily through reading, listening to texts read aloud, and engaging with educational media — creating a cycle where students with fewer of these experiences fall further behind in vocabulary.

- For English language learners, core vocabulary can present an additional challenge not because of any difference in ability, but because of reduced opportunity for exposure. These words appear almost exclusively in academic settings and written texts, contexts that English learners may have had less access to.

- Many academic words have meanings that are more abstract or technical than everyday words. A word like table may seem familiar but be used in ways that are subtly different from casual speech (for example, in everyday conversation table might mean the object where you eat dinner, but in a political science text it often signals a delay in discussion, like when someone says “let’s table the discussion”)

- Students may recognize a word without having a reliable, accessible meaning for it. They have seen it before but cannot use it fluently during reading. This partial knowledge can mask a genuine gap.

HOW DO READERS DEVELOP THIS SKILL?

Vocabulary knowledge grows through two complementary routes: wide exposure to language-rich input (e.g. reading, listening to texts read aloud, and engaging with educational media) which builds incidental exposure to words across many contexts, and explicit instruction, which directly teaches the meanings of high-priority core vocabulary words. Both routes are needed: wide exposure builds familiarity and enables students to learn new words independently over time, while explicit instruction of carefully selected words deepens that knowledge and accelerates growth. Research consistently shows that explicit instruction in core vocabulary, particularly when it involves multiple encounters with words across varied contexts, produces the strongest gains in both vocabulary and comprehension.

Students benefit most from instruction that goes beyond simple definitions. Teaching a word means helping students understand how it relates to other words they know, how it functions in sentences, and how to recognize it across different contexts and forms. Repeated, meaningful encounters with academic vocabulary, in reading, discussion, and writing, helps students move from partial familiarity to the kind of reliable, fluent knowledge that supports comprehension.

A student who knows that *analyze* means to examine something carefully and break it into parts can apply that knowledge in science lab reports, history essays, and math problem explanations, the same word doing the same work across every subject.

NEXT STEPS

Explicit, vocabulary instruction with multiple exposures across varied contexts produces the strongest gains. The following programs and resources support academic vocabulary development across grades:

- TextProject — free resources grounded in the core vocabulary research that underlies ROAR-Written Vocabulary, including word lists and instructional materials

- NCIL Explicit Vocabulary Instruction — a brief and infographic grounded in the science of reading with practical guidance on what and how to teach vocabulary

- Reading Universe Vocabulary — videos, tools, and classroom resources for teaching word meanings and relationships

- Meadows Center 10 Key Vocabulary Practices — research-based recommendations for vocabulary instruction across grade levels

- Achieve the Core: Selecting and Using Academic Vocabulary — guidance and classroom activities for selecting and teaching high-value vocabulary words

ROAR-MORPHOLOGY

MORPHOLOGICAL KNOWLEDGE

WHAT IS IT?

Morphological knowledge is the understanding of meaningful units within words, units called *morphemes*, and how they combine to convey meaning. Morphemes are the smallest units of meaning in a language. They include base words (like *help*), prefixes (like *un*-), and suffixes (like -*ful* or -*ed*). A reader with strong morphological knowledge can look at an unfamiliar word like unhelpful and recognize that it is built from parts they already know.

Morphological knowledge is closely related to vocabulary knowledge, reading comprehension, and spelling, making it a key bridge between foundational decoding skills and higher-level reading.

WHY IS IT IMPORTANT?

As students progress through school, the words they encounter in academic texts become longer and more complex. Research shows that the majority of new academic vocabulary words are morphologically complex, meaning they contain prefixes, suffixes, or Latin and Greek roots that carry meaning across many words. A student who can break apart a word like *microscopic* into *micro*- (small), -*scop*- (see), and -*ic* (relating to) has a powerful tool for figuring out unfamiliar words across content areas.

Morphological knowledge supports reading development in at least three ways: it helps students decode multisyllabic words more efficiently, it builds vocabulary breadth and depth, and it improves reading comprehension. This form of knowledge is important for the academic and disciplinary texts students encounter in the upper elementary grades and beyond.

WHAT ARE THE COMMON CHALLENGES WITH THIS SKILL?

- Students may recognize a base word in isolation but struggle to identify it when it is embedded in a longer, morphologically complex word (e.g., knowing *sign* but not connecting it to *signature* or *signal*).

- Many morphological relationships in English are partly hidden — the spelling stays consistent even when pronunciation changes (e.g., *heal* → *health*, *please* → *pleasant*). These *opaque* relationships require explicit instruction to recognize.

- Students who are still working on decoding may not have the attentional resources to notice word structure. For these students, decoding should remain the primary instructional focus, with morphology introduced as a tool for supporting that work — for example, using knowledge of the prefix *re*- and base word *build* to decode *rebuilding.*

- English most commonly draws on Latin, Greek, and Anglo-Saxon root systems, each with notably different patterns. Students benefit from instruction that builds this knowledge systematically rather than word by word.

HOW DO READERS DEVELOP THIS SKILL?

Morphological knowledge develops gradually across the school years. Young children begin to understand simple inflections (like -s for plural or -ed for past tense) through oral language. As they become readers, they start to notice morphological patterns in print, but this knowledge deepens with explicit instruction.

Direct teaching of high-utility prefixes, suffixes, and roots is an effective approach. Instruction that helps students actively analyze word structure, make connections between related words, and apply morphological knowledge while reading produces the strongest outcomes. Like phonics, morphology instruction is most effective when it is systematic and builds from simpler to more complex patterns.

Teaching students that the prefix re- means “again” helps them unlock not just reread, but also rewrite, rebuild, reconsider, and dozens of other words they will encounter across subjects.

NEXT STEPS

Morphological knowledge is best developed through explicit, systematic instruction that connects word structure to meaning. The following programs and resources integrate morphology into vocabulary and reading comprehension instruction:

For all grades:

- Florida Center for Reading Research (fcrr.org) — free classroom activities targeting morphological awareness and word structure.
- Reading Universe — practitioner resources on morphology and word meaning instruction.
- Shanahan on Literacy — evidence-based overview of what morphology instruction should look like.
- Mini Matrix-Maker — a free interactive tool for building word matrices to explore how bases, prefixes, and suffixes combine to form word families.
- Morpheme Match Game - a free lesson plan on morpheme identification and use intended for grades 3-5.

ROAR-INFERENCE

INFERENTIAL REASONING

WHAT IS IT?

Language comprehension, the ability to construct meaning from what we read and hear, requires more than understanding individual sentences. Readers must constantly fill in what is not explicitly stated, connecting ideas within a text and drawing on their own knowledge to construct meaning. This process is called *inferential reasoning,* and it is central to how skilled readers make sense of what they read.

ROAR-Inference measures students' ability to evaluate and select coherent explanations from short passages. After reading a 2 to 6 sentence passage, students answer questions by selecting the response that best explains what they read. The response options are not simply right or wrong: the answer choices are designed to suggest how students are thinking. One answer reflects full understanding of the passage; one shows a student noticed something relevant but didn't connect it to the bigger picture; one reflects a guess based on general knowledge rather than what the text actually says.

The assessment covers three types of meaning-making that readers engage in when processing narrative text:

INFERENCE TYPES

LOGICAL	INFORMATIONAL	EVALUATIVE
Understanding causality and motivation. Why did a character act this way? How are events connected?	Tracking who is involved, what happened, where, and when. How are details connected within the story?	Understanding broader significance. What does this reveal about a character? What is the lesson or theme?

Students read the following passage and are asked "*What does this passage teach us?*"

> *"The team worked hard all season. Their coach led them to victory and won the biggest prize. She decided to share the prize with every member of the team."*

- A student who answers "*Sharing what you have with others brings joy*" is pulling together multiple ideas from the text into a coherent lesson.
- A student who answers "*How to win a prize*" has noticed a detail from the passage but hasn't connected it to the larger meaning.

- A student who answers "*She wanted to become famous*" is drawing on general knowledge about why people might share prizes — it's a plausible real-world reason — but nothing in the passage supports this interpretation. This response reveals a student who is relying on background knowledge rather than grounding their answer in the text.

The difference reveals how deeply the student is engaging with what the text is really about.

WHY IS IT IMPORTANT?

Students who struggle with inferential reasoning may understand individual sentences perfectly well but fail to construct a coherent picture of a passage as a whole. They can repeat back surface details but miss the connections that give those details meaning. This difficulty often becomes more consequential as texts grow longer and more complex across the grades, creating a comprehension bottleneck that is hard to spot from the outside.

Inferential comprehension draws on different types of meaning-making, such as understanding causality, tracking informational elements, and recognizing deeper significance, and these work together as an integrated skill. ROAR-Inference captures this as a single overall score. Within that score, the pattern of answer choices students make can still reveal something specific about where their meaning-making tends to break down. For example, whether they over-rely on background knowledge, focus on surface details, or struggle to connect ideas into a coherent explanation. Beyond the type of meaning-making required, questions also vary in how directly the answer can be found in the text, a distinction captured by a framework reading teachers often use called Question-Answer Relationships (QAR): some answers are right in the text; others require students to think and search across the passage; and others questions combine text with background knowledge. This variation is a key reason why students may find some questions easier than others.

WHAT ARE THE COMMON CHALLENGES WITH THIS SKILL?

- Some students over-rely on background knowledge, constructing explanations that seem plausible but are not grounded in what the text says. They answer from general world knowledge rather than from the passage itself.

- Other students focus on surface-level details, identifying something true from the text without connecting it to the specific question being asked. They access textual information but do not integrate it into a coherent explanation.

- Understanding causal and motivational relationships is particularly demanding, because these connections are rarely spelled out explicitly. Readers must infer why a character acted a certain way or how one event led to another.

- Informational meaning-making involves tracking who is involved, what happened, where, and when. It can be challenging when passages contain multiple characters, objects, or events that students must keep track of simultaneously.

- Evaluative meaning-making, identifying what events reveal about a character's values or the broader theme of a passage, requires students to move beyond what happened in the text and consider what it means.

- Students with comprehension difficulties often show adequate decoding skills, meaning their inferencing challenges can go unnoticed until texts become demanding enough to expose the gap.

HOW DO READERS DEVELOP THIS SKILL?

Children begin constructing meaning from events and narratives from a very young age. They infer causes, track who did what, and draw lessons from stories well before they can read. As students develop as readers, this capacity becomes increasingly sophisticated: they learn to check their explanations against textual evidence, integrate multiple pieces of information, and evaluate competing interpretations. Stronger readers also develop the ability to inhibit less relevant information, setting aside plausible but unsupported interpretations in favor of explanations that are actually grounded in the text.

Explicit instruction in critically analysing and questioning text supports this development. Teaching students to ask why and how questions as they read, to identify which details are essential rather than incidental, and to evaluate their own interpretations against the text produces meaningful gains in inferential comprehension. Instruction is most effective when it is embedded in real reading rather than practiced in isolation and when it gives students opportunities to discuss and compare their reasoning with others. Discussion is especially powerful when students talk through their interpretations, explain their reasoning, and hear how classmates read the same passage differently. Developing inferencing skills through discussion helps to build the habit of checking their understanding against the text, which is exactly what strong inferential readers do.

NEXT STEPS

Inferential comprehension responds to explicit strategy instruction embedded in real reading. The following programs and resources support inference development across grades:

For grades K–8:
- Reciprocal Teaching (Palincsar & Brown) — a well-evidenced comprehension strategy integrating inference, prediction, clarification, and summarizing.

For grades 2–8:
- Questioning the Author (Beck & McKeown) — a structured approach to building meaning from text through collaborative discussion and questioning.

For all grades:
- Reading Rockets — teacher-friendly overview of inference instruction strategies, practical guides, and classroom activities.

- Florida Center for Reading Research — free comprehension activities targeting inference, cause-and-effect reasoning, and drawing conclusions.

- IES Practice Guides — evidence-based recommendations for reading comprehension instruction.

END NOTES

Our assessments of these skills have been validated against gold standard assessments in phonemic awareness (CTOPP), word recognition (Woodcock-Johnson), and sentence reading efficiency (TOSREC). We are currently running studies to validate the assessment as a dyslexia screener specifically. The linked research shows that our assessment provides similar information to other gold-standard foundational reading skill assessments.

Further, we have validated the Computer Adaptive Testing algorithm we use in ROAR-Word, and our methods for generating and testing new items for sentence-reading efficiency. To see an overview of research and development, please refer to our ROAR Technical Manual.